GECG10069 (561084) F26: Introduction to Programming (C++)

Lab 5 : Literals & Variables



What you will learn from Lab 5

In this laboratory, you will further explore special output symbols and practice using variables in your programs.

TASK 5-1 : IF V.S. ELSE IF

✔ **Version 1 (Correct with else if)**

- Conditions are mutually exclusive.
- Once one condition is true, the others are skipped.
- Clear and concise, no need to repeat conditions.

✔ **Version 2 (Wrong with only if)**

- Multiple conditions may be true at the same time.
- Example: input 95 prints A, B, C.
- Causes wrong or duplicated outputs.

✔ **Version 3 (Correct but redundant)**

- Each if repeats the full range check.
- Works correctly but is verbose and harder to maintain.

✔ **Don't forget braces!!**

- In this example, each if or else if has only a single statement, so braces are optional.
- If you have multiple statements inside the block, always use braces { } to avoid logic errors and improve readability.

```
#include <iostream>
using namespace std;

int main()
{
    int ss;
    cin >> ss; //ss is int
    // Correct
    cout << "Version 1: ";
```

```
  if (ss>=90)     cout << "A" << endl; //ss≥90
  else if (ss>=80) cout << "B" << endl; //90>ss≥80
  else if (ss>=70) cout << "C" << endl; //80>ss≥70

    // Wrong
    cout << "Version 2: ";
  if (ss>=90) cout << "A" << endl; //ss≥90
  if (ss>=80) cout << "B" << endl; //90>ss≥80
  if (ss>=70) cout << "C" << endl; //80>ss≥70

    // Correct but redundant
    cout << "Version 3: ";
  if (ss>=90)       cout << "A" << endl; //ss≥90
  if (ss>=80 && ss<90) cout << "B" << endl; //90>ss≥80
  if (ss>=70 && ss<80) cout << "C" << endl; //80>ss≥70

    return 0;
}
```

TASK 5-2 : SWITCH

- ✔ **With break (Correct):**
  - Each case ends properly, execution stops after matching case.
  - Example: input 1 → only prints Select 1.
- ✔ **Without break (Wrong):**
  - Execution "falls through" into the next cases.
  - Example: input 1 → prints Select 1, Select 2, and Error.
  - This is usually unintended and causes wrong outputs.
- ✔ **Don't forget break!!**
  - In this example, each if or else if has only a single statement, so braces are optional.
  - If you have multiple statements inside the block, always use braces { } to avoid logic errors and improve readability.

```
#include <iostream>
using namespace std;

int main()
{
```

```
    int menu;
    cin >> menu;
    switch (menu) {
        // Corect
        case 1: cout << "Select 1\n"; break;
        case 2: cout << "Select 2\n"; break;
        default: cout << "Error\n";

        // Wrong
        // case 1: cout << "Select 1\n";
        // case 2: cout << "Select 2\n";
        // default: cout << "Error\n";
    }
    return 0;
}
```

EXERCISE 5-1: EQUATION CHECKER

Write a C++ program that reads a simple equation from a file and decides whether it is correct.

Requirements

- You are given a file named data.txt containing **five** equation in the form

  $$a\ op\ b = c$$

  where a, b and c are integers and op is one of +, -, *, /. The parts are separated by spaces.

- **Print yes with cout if the equation is correct, otherwise no.**
- **a, b, c are all integers.**
- **a and b will not be 0.**

Sample Input	Sample Output
7 + 2 = 9	yes
1 + 1 = 3	no
9 - 4 = 6	no
6 * 7 = 42	yes
36 / 6 = 12	no

EXERCISE 5-2: INFO PRINTER

Write a C++ program to print out your information based on different input modes.

Requirements

- Input: first a power command, read with cin. It is on or off, in lowercase.
- If the power is off, print Standby. **There is no mode command to read.**
- If the power is on, read a second command, the mode. It is 1, 2, or 3:
  - mode 1 — print your **name**
  - mode 2 — print your **department,** abbreviation is okay, like EE.
  - mode 3 — print your **student number**
- Print one line of output with cout.

Sample Input - 1
on 1
Sample Output - 1
Nai Long
Sample Input - 2
on 3
Sample Output - 2
115114514
Sample Input - 3
off
Sample Output - 3
Standby